We create chemistry

# Safety data sheet



BASF safety data sheet. This is a translation of the country-specific safety data sheet into a language other than that required by law. This document does not replace the safety data sheet provided according to Regulation (EC) No 1907/2006.

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Date previous version: 25.01.2023 Previous version: 2.0
Date / First version: 13.09.2022
Product: **CYCLOHEXAN LU**

(ID no. 30678576/SDS_GEN_DE/EN)

Date of print 06.07.2025

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

# CYCLOHEXAN LU

Chemical name: cyclohexane
CAS Number: 110-82-7

REACH registration number: 01-2119463273-41-0010, 01-2119463273-41-0012, 01-2119463273-41-0023, 01-2119463273-41

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended use: Chemical

For the detailed identified uses of the product see appendix of the safety data sheet.

### 1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Petrochemicals

Telephone: +49 621 60-42151
E-mail address: sds-petrochemicals@basf.com

### 1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

## SECTION 2: Hazards Identification

### 2.1. Classification of the substance or mixture

According to Regulation (EC) No 1272/2008 [CLP]

| | |
|---|---|
| Asp. Tox. 1 | H304 May be fatal if swallowed and enters airways. |
| Flam. Liq. 2 | H225 Highly flammable liquid and vapour. |
| Skin Corr./Irrit. 2 | H315 Causes skin irritation. |
| STOT SE 3 | H336 May cause drowsiness or dizziness. |
| Aquatic Acute 1 | H400 Very toxic to aquatic life. |
| Aquatic Chronic 1 | H410 Very toxic to aquatic life with long lasting effects. |

M-factor acute: 1

M-factor chronic: 1

For the classifications not written out in full in this section the full text can be found in section 16.

### 2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Danger

Hazard Statement:

| | |
|---|---|
| H225 | Highly flammable liquid and vapour. |
| H315 | Causes skin irritation. |
| H304 | May be fatal if swallowed and enters airways. |
| H336 | May cause drowsiness or dizziness. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear protective gloves and eye protection or face protection. |
| P210 | Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. |
| P273 | Avoid release to the environment. |

Precautionary Statements (Response):

| | |
|---|---|
| P312 | Call a POISON CENTER or physician if you feel unwell. |

Precautionary Statements (Storage):

| | |
|---|---|
| P233 | Keep container tightly closed. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

### 2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
See section 12 - Results of PBT and vPvB assessment.
Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.
The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative).

## SECTION 3: Composition/Information on Ingredients

### 3.1. Substances

Chemical nature

cyclohexane

| | |
|---|---|
| Content (W/W): >= 99,9 % | Asp. Tox. 1 |
| CAS Number: 110-82-7 | Flam. Liq. 2 |
| EC-Number: 203-806-2 | Skin Corr./Irrit. 2 |
| INDEX-Number: 601-017-00-1 | STOT SE 3 (drowsiness and dizziness) |
| | Aquatic Acute 1 |
| | Aquatic Chronic 1 |
| | M-factor acute: 1 |
| | M-factor chronic: 1 |
| | H225, H315, H304, H336, H400, H410 |

Regulatory relevant ingredients

cyclohexane

| | |
|---|---|
| Content (W/W): >= 99,9 % - <= 99,99 % | Asp. Tox. 1 |
| | Flam. Liq. 2 |
| CAS Number: 110-82-7 | Skin Corr./Irrit. 2 |
| EC-Number: 203-806-2 | STOT SE 3 (drowsiness and dizziness) |
| INDEX-Number: 601-017-00-1 | Aquatic Acute 1 |
| | Aquatic Chronic 1 |
| | M-factor acute: 1 |
| | M-factor chronic: 1 |
| | H225, H315, H304, H336, H400, H410 |

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

### 3.2. Mixtures

Not applicable

## SECTION 4: First-Aid Measures

### 4.1. Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

### 4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Hazards: When inhaled (e.g. during vomiting) risk of pulmonary oedema and/or pneumonia.

### 4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## SECTION 5: Fire-Fighting Measures

### 5.1. Extinguishing media

Suitable extinguishing media:
dry powder, water spray, carbon dioxide, foam

Unsuitable extinguishing media for safety reasons:
water jet

Additional information:
Use extinguishing measures to suit surroundings.

### 5.2. Special hazards arising from the substance or mixture

Advice: Highly flammable. Cool endangered containers with water-spray. See SDS section 7 - Handling and storage.

### 5.3. Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus. Special protective equipment for firefighters

Further information:
Evacuate area of all unnecessary personnel. Fight fire from maximum distance.

Extend fire extinguishing measures to the surroundings. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## SECTION 6: Accidental Release Measures

High risk of slipping due to leakage/spillage of product.

Release of substance/product can cause fire or explosion. Shut off or stop source of leak. Shut off or stop released substance/product under safe conditions.

Pack in tightly closed containers for disposal.

### 6.1. Personal precautions, protective equipment and emergency procedures

Handle in accordance with good industrial hygiene and safety practice.

Avoid all sources of ignition: heat, sparks, open flame. Use antistatic tools. Avoid contact with the skin, eyes and clothing.

Take off immediately all contaminated clothing.

### 6.2. Environmental precautions

Due to the pH-value of the product, neutralization is generally required before discharging sewage into treatment plants.

Discharge into the environment must be avoided.

### 6.3. Methods and material for containment and cleaning up

Pick up with suitable appliance and dispose of. Spills should be contained, solidified, and placed in suitable containers for disposal. Dispose of absorbed material in accordance with regulations.

### 6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

## SECTION 7: Handling and Storage

### 7.1. Precautions for safe handling

Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:
Avoid all sources of ignition: heat, sparks, open flame. Ground all transfer equipment properly to prevent electrostatic discharge.

Electrical devices must meet the specified temperature class.

Temperature class: T3 (Autoignition temperature >200 °C).

### 7.2. Conditions for safe storage, including any incompatibilities

No applicable information available.
Further information on storage conditions: Keep container tightly closed in a cool, well-ventilated place.

Storage class according to TRGS 510 (originally VCI, Germany): (3) Flammable liquids

### 7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

## SECTION 8: Exposure Controls/Personal Protection

### 8.1. Control parameters

Components with occupational exposure limits

The surveillance of the workplace by exposure measurements may be necessary, in order to prove the efficiency of safety measures, for example ventilation or the need of respiratory protection. Since this requires a specific competency, only accredited laboratories should be contracted. Regarding suitable methods to assess inhalation exposure, the European Standards EN 482, 689 and 14042 are to be considered. In addition, the TRGS 402 has to be observed in Germany.

110-82-7: cyclohexane

TWA value 700 mg/m3 ; 200 ppm (OEL (EU))
indicative
OEL 700 mg/m3 ; 200 ppm (TRGS 900 (DE))
Ceiling limit value/factor: 4
Short Term Exposure Classification: (TRGS 900 (DE))
Category II: Substances with a resorptive effect

Biological limit values (BLV)

TRGS 903 (DE)
Determinant: 1,2-Cyclohexanediol, with hydrolysis
Biological Specimen: Creatinine in urine
Sampling time: End of shift at end of work week
Concentration: 150 mg/g Creatinine

PNEC

freshwater: 0,207 mg/l

marine water: 0,207 mg/l

intermittent release: 0,207 mg/l

sediment (freshwater): 3,627 mg/kg

sediment (marine water): 3,627 mg/kg

soil: 2,99 mg/kg

STP: 3,24 mg/l

DNEL

worker:
Long-term exposure- systemic effects, Inhalation: 700 mg/m3, 200 ppm

worker:
Short-term exposure - systemic effects, Inhalation: 700 mg/m3, 200 ppm

worker:
Long-term exposure - local effects, Inhalation: 700 mg/m3, 200 ppm

worker:
Short-term exposure - local effects, Inhalation: 700 mg/m3, 200 ppm

worker:
Long-term exposure- systemic effects, dermal: 2016 mg/kg

consumer:
Short-term exposure - systemic effects, Inhalation: 412 mg/m3

consumer:
Long-term exposure - systemic and local effects, Inhalation: 206 mg/m3

consumer:
Short-term exposure - local effects, Inhalation: 412 mg/m3

consumer:
Long-term exposure - local effects, Inhalation: 206 mg/m3

consumer:
Long-term exposure- systemic effects, dermal: 1186 mg/kg

consumer:
Long-term exposure- systemic effects, oral: 59,4 mg/kg

## 8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Wear respiratory protection if ventilation is inadequate. Gas filter for gases/vapours of organic compounds (boiling point >65 °C, e. g. EN 14387 Type A)

Hand protection:
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
fluoroelastomer (FKM) - 0.7 mm coating thickness
nitrile rubber (NBR) - 0.4 mm coating thickness

Suitable materials for short-term contact (recommended: At least protective index 2, corresponding > 30 minutes of permeation time according to EN ISO 374-1)
chloroprene rubber (CR) - 0.5 mm coating thickness
butyl rubber (butyl) - 0.7 mm coating thickness

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Wearing of closed work clothing is required additionally to the stated personal protection equipment. Handle in accordance with good industrial hygiene and safety practice. At the end of the shift the skin should be cleaned and skin-care agents applied.

Environmental exposure controls

All appropriate measures must be taken to prevent the release of this product to the environment and to limit the dispersion of any release when it occurs. Suitable risk management measures should be in place.

## SECTION 9: Physical and Chemical Properties

### 9.1. Information on basic physical and chemical properties

| | | |
|---|---|---|
| State of matter: | liquid | |
| Form: | liquid | |
| Colour: | colourless, clear | |
| Odour: | faintly aromatic | |
| Odour threshold: | not determined | |
| Melting point: | 6,5 °C<br>Literature data. | |
| Boiling point: | 80,7 °C<br>(1.013 hPa)<br>Literature data. | |
| Flammability: | Highly flammable. | (derived from flash point) |
| Lower explosion limit: | 1,3 %(V)<br>(20 °C, 1 bar)<br><br>The lower explosion point may be 5 - 15 °C below the flash point., For liquids not relevant for classification and labelling. | (air) |
| Upper explosion limit: | 8,4 %(V)<br>(20 °C, 1 bar) | (air) |

| | | |
|---|---|---|
| | For liquids not relevant for classification and labelling. | |
| Flash point: | -20 °C<br>Literature data. | (closed cup) |
| Auto-ignition temperature: | 260 °C<br>Literature data. | |
| Thermal decomposition: | No decomposition if correctly stored and handled. | |
| pH value: | The substance does not dissociate. | |
| Viscosity, kinematic: | 1,155 mm2/s<br>(20 °C) | |
| Viscosity, dynamic: | 0,894 mPa.s<br>(20 °C) | |
| Thixotropy: | not thixotropic | |
| Solubility in water: | Literature data.<br>0,052 g/l<br>(23,5 °C) | |
| Partitioning coefficient n-octanol/water (log Kow): | 3,44<br>(25 °C; pH value: 7) | (calculated) |
| Vapour pressure: | 124 hPa<br>(24 °C)<br>Literature data. | (measured) |
| Relative density: | 0,7739<br>(20 °C) | |
| Density: | 0,7739 g/cm3<br>(20 °C)<br>Literature data. | |
| Relative vapour density (air): | > 1<br>(20 °C)<br>Heavier than air. | (estimated) |

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

## 9.2. Other information

**Information with regard to physical hazard classes**

Explosives

Explosion hazard: Based on the chemical structure there is no indication of explosive properties.

Impact sensitivity: Based on the chemical structure there is no shock-sensitivity.

Oxidizing properties

Fire promoting properties: not fire-propagating

Pyrophoric properties

| | | |
|---|---|---|
| Self-ignition temperature: | | Test type: Spontaneous self-ignition at room-temperature. |
| | Based on its structural properties the product is not classified as self-igniting. | |

Self-heating substances and mixtures

| | |
|---|---|
| Self heating ability: | It is not a substance capable of spontaneous heating. |

Substances and mixtures, which emit flammable gases in contact with water

Formation of flammable gases:
Forms no flammable gases in the presence of water.

Corrosion to metals

No corrosive effect on metal.

**Other safety characteristics**

| | | |
|---|---|---|
| Radioactivity: | | not radioactive for transport purposes |
| pKA: | The substance does not dissociate. | |
| Adsorption/water - soil: | KOC: 770; log KOC: 2,89 | (calculated) |
| Surface tension: | Based on chemical structure, surface activity is not to be expected. | |
| Molar mass: | 84,16 g/mol | |
| Evaporation rate: | Value can be approximated from Henry's Law Constant or vapor pressure. | |

## SECTION 10: Stability and Reactivity

### 10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated., When heated can give off ignitable vapours., Vapours may form explosive mixture with air.

| | | |
|---|---|---|
| Corrosion to metals: | No corrosive effect on metal. | |
| Formation of flammable gases: | Remarks: | Forms no flammable gases in the presence of water. |

### 10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Peroxides: The product/the substance has not a tendency towards the formation of peroxide.

### 10.3. Possibility of hazardous reactions

No hazardous reactions if stored and handled as prescribed/indicated.

### 10.4. Conditions to avoid

Avoid all sources of ignition: heat, sparks, open flame.

### 10.5. Incompatible materials

Substances to avoid:
strong oxidizing agents

### 10.6. Hazardous decomposition products

No hazardous decomposition products if stored and handled as prescribed/indicated.

## SECTION 11: Toxicological Information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Assessment of acute toxicity:

Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact. Virtually nontoxic after a single ingestion.

Experimental/calculated data:

LD50 rat (oral): > 5.000 mg/kg (similar to OECD guideline 401)
No mortality was observed. Limit concentration test only (LIMIT test).
LC50 rat (by inhalation): > 32,88 mg/l 4 h (similar to OECD guideline 403)
No mortality was observed. The vapour was tested.
LD50 rabbit (dermal): > 2.000 mg/kg (similar to OECD guideline 402)
No mortality was observed. Limit concentration test only (LIMIT test).

Irritation

Assessment of irritating effects:

Skin contact causes irritation. EU-classification Not irritating to the eyes.

Experimental/calculated data:

Skin corrosion/irritation

rabbit: Slightly irritating. (similar to OECD guideline 404)
The European Union (EU) has classified this substance with 'Irritating to skin' (R38).

Serious eye damage/irritation

rabbit: non-irritant (similar to OECD guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:

Skin sensitizing effects were not observed in animal studies.

Experimental/calculated data:

Buehler test guinea pig: Non-sensitizing. (other)

Germ cell mutagenicity

Assessment of mutagenicity:

No mutagenic effect was found in various tests with bacteria and mammalian cell culture. The substance was not mutagenic in a test with mammals.

Carcinogenicity

Assessment of carcinogenicity:
The whole of the information assessable provides no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:

No indications of a developmental toxic / teratogenic effect were seen in animal studies.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Possible narcotic effects (drowsiness or dizziness).

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Repeated exposure to the substance by inhalative administration leads to effects similar to those found after single exposure.

Aspiration hazard

May also damage the lung at swallowing (aspiration hazard).

Interactive effects

No data available.

### 11.2. Information on other hazards

Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

## SECTION 12: Ecological Information

### 12.1. Toxicity

Assessment of aquatic toxicity:
Very toxic (acute effect) to aquatic organisms. Depending on local conditions and existing concentrations, disturbances in the biodegradation process of activated sludge are possible.

Toxicity to fish:
LC50 (96 h) 4,53 mg/l, Pimephales promelas (Fish test acute, Flow through.)
The statement of the toxic effect relates to the analytically determined concentration.

Aquatic invertebrates:
EC50 (48 h) 0,9 mg/l, Daphnia magna (Daphnia test acute, static)
The statement of the toxic effect relates to the analytically determined concentration.

Aquatic plants:
EC50 (72 h) 9,3 mg/l (growth rate), Selenastrum capricornutum (OECD Guideline 201)
The statement of the toxic effect relates to the analytically determined concentration.

No observed effect concentration (72 h) 0,94 mg/l (growth rate), Selenastrum capricornutum (OECD Guideline 201)
The statement of the toxic effect relates to the analytically determined concentration.

Microorganisms/Effect on activated sludge:
EC50 (15 h) 29 mg/l, aerobic microorganisms (Oxygen consumption test, aquatic)

Chronic toxicity to fish:
No data available regarding toxicity to fish.

Chronic toxicity to aquatic invertebrates:
No data available regarding toxicity to daphnids.

Soil living organisms:
LC50 (48 h) > 1 mg/cm2, Eisenia foetida (Orientating study, filter paper)

## 12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Readily biodegradable (according to OECD criteria).

Elimination information:
77 % BOD of the ThOD (28 d) (OECD Guideline 301 F) (aerobic, activated sludge, domestic, non-adapted)

Assessment of stability in water:
According to structural properties, hydrolysis is not expected/probable.

## 12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

Bioaccumulation potential:

Bioconcentration factor(BCF): 167, Pimephales promelas (calculated)
Literature data.

## 12.4. Mobility in soil

Assessment transport between environmental compartments:
Volatility: The substance will rapidly evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

## 12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative). Self classification

## 12.6. Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

## 12.7. Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

### 12.8. Additional information

Other ecotoxicological advice:
Do not release untreated into natural waters. Very toxic (acute effect) to aquatic organisms.

## SECTION 13: Disposal Considerations

### 13.1. Waste treatment methods

Dispose of in accordance with national, state and local regulations.

Contaminated packaging:
Disposal must be made according to official regulations.

## SECTION 14: Transport Information

**Land transport**

ADR

| | |
|---|---|
| UN number or ID number: | UN1145 |
| UN proper shipping name: | CYCLOHEXANE |
| Transport hazard class(es): | 3, EHSM |
| Packing group: | II |
| Environmental hazards: | yes |
| Special precautions for user: | Tunnel code: D/E |

RID

| | |
|---|---|
| UN number or ID number: | UN1145 |
| UN proper shipping name: | CYCLOHEXANE |
| Transport hazard class(es): | 3, EHSM |
| Packing group: | II |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| UN number or ID number: | UN1145 |
| UN proper shipping name: | CYCLOHEXANE |
| Transport hazard class(es): | 3, EHSM |
| Packing group: | II |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

Transport in inland waterway vessel

| | |
|---|---|
| UN number or ID number: | UN1145 |
| UN proper shipping name: | CYCLOHEXANE |
| Transport hazard class(es): | 3, N1 |
| Packing group: | II |
| Environmental hazards: | yes |
| Type of inland waterway vessel: | C |
| Cargo tank design: | 2 |
| Cargo tank type: | 2 |

**Sea transport**

IMDG

| | |
|---|---|
| UN number or ID number: | UN 1145 |
| UN proper shipping name: | CYCLOHEXANE |
| Transport hazard class(es): | 3, EHSM |
| Packing group: | II |
| Environmental hazards: | yes<br>Marine pollutant: YES |
| Special precautions for user: | EmS: F-E; S-D |

**Air transport**

IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 1145 |
| UN proper shipping name: | CYCLOHEXANE |
| Transport hazard class(es): | 3 |

| | |
|---|---|
| Packing group: | II |
| Environmental hazards: | No Mark as dangerous for the environment is needed |
| Special precautions for user: | None known |

### 14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

### 14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

### 14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

### 14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

### 14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

### 14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

### 14.7. Maritime transport in bulk according to IMO instruments

| | |
|---|---|
| Regulation: | IBC-Code |
| Product name: | Cyclohexane |
| Pollution category: | Y |
| Ship Type: | 2 |

## SECTION 15: Regulatory Information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Prohibitions, Restrictions and Authorizations

Annex XVII of Regulation (EC) No 1907/2006: Number on List: 57, 3, 40

Hazardous Incident Ordinance (Germany):
List entry in regulation: 1.2.5.2
List entry in regulation: 1.3.1

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
List entry in regulation: E1
List entry in regulation: P5b

Classification according to 'TA-Luft' (Germany):

5.2.5: Organic gases, general guidance

Water hazard class (§6 AwSV para.4 (Legal binding announcement of the substance in the Federal Gazette)): (2) significantly water polluting. ID-No.: 63

German Regulation TA Luft (Technical Instruction on Air Quality Control, i.e. first Directive to the Federal Immission Control Ordinance)
The specifications of the Technical Rule for Hazardous Substances (TRGS) 401 must be observed (TRGS 401: Risks resulting from skin contact - identification, assessment, measures).
Law on the Protection of Working Youth

### 15.2. Chemical Safety Assessment

Chemical Safety Assessment performed

## SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Asp. Tox. 1
STOT SE 3 (Vapours may cause drowsiness and dizziness.)
Aquatic Acute 1
Flam. Liq. 2
Aquatic Chronic 1
Skin Corr./Irrit. 2

M-factor acute: 1
M-factor chronic: 1

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Asp. Tox. | Aspiration hazard |
| Flam. Liq. | Flammable liquids |
| Skin Corr./Irrit. | Skin corrosion/irritation |
| STOT SE | Specific target organ toxicity — single exposure |

| | |
|---|---|
| Aquatic Acute | Hazardous to the aquatic environment - acute |
| Aquatic Chronic | Hazardous to the aquatic environment - chronic |
| H225 | Highly flammable liquid and vapour. |
| H315 | Causes skin irritation. |
| H304 | May be fatal if swallowed and enters airways. |
| H336 | May cause drowsiness or dizziness. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

Abbreviations

ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

## Annex: Exposure Scenarios

**Index**

**1.**
ERC1; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC28, PROC8b, PROC9, PROC15

**2.**
ERC2; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC28, PROC8b, PROC9, PROC14, PROC15

**3.**
ERC6a; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC28, PROC8b, PROC9

**4.**
ERC4; PROC1, PROC2, PROC3, PROC4, PROC5, PROC7, PROC8a, PROC28, PROC8b, PROC9, PROC10, PROC13, PROC14, PROC15

**5.**
ERC4; PROC1, PROC2, PROC3, PROC4, PROC7, PROC8a, PROC8b, PROC10, PROC13

**6.**
ERC4; PROC1, PROC2, PROC3, PROC8b, PROC9, PROC12

**7.**
ERC4; PROC1, PROC2, PROC3, PROC4, PROC6, PROC7, PROC8a, PROC28, PROC8b, PROC10, PROC13, PROC14

**8.**
ERC7; PROC1, PROC2, PROC3, PROC8a, PROC28, PROC8b, PROC16

**9.**
ERC6c; PROC1, PROC2, PROC3, PROC4, PROC8b, PROC21

**10.**
ERC4; PROC1, PROC2, PROC3, PROC4, PROC5, PROC6, PROC8a, PROC28, PROC8b, PROC13, PROC14, PROC21

**11.**
ERC4; PROC15

**12.**
ERC8d; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC28, PROC8b, PROC10, PROC11, PROC13, PROC15, PROC19

**13.**
ERC8d; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC10, PROC11, PROC13, PROC19

**14.**

ERC9b; PROC1, PROC2, PROC3, PROC8a, PROC28, PROC16

**15.**
ERC8a; PROC15

**16.**
ERC8d; PROC1, PROC2, PROC6, PROC21, PROC8a, PROC28, PROC8b

**17.**
ERC9b; PC13

**18.**
ERC8d; PC1, PC9a, PC15, PC18, PC24, PC34

**********

## 1. Short title of exposure scenario
ERC1; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC28, PROC8b, PROC9, PROC15

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC1: Manufacture of the substance |
| ***Operational conditions*** | |
| Annual amount per site | 150.000.000 kg |
| Minimum emission days per year | 300 |
| Emission factor air | 0,5 % |
| Emission factor water | 0,003 % |
| Emission factor soil | 0,01 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 40 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,604869 |

| | |
|---|---|
| | Risk from environmental exposure is driven by soil. |
| Maximum amount of safe use | 826.624,9<br>kg/d |
| Risk from environmental exposure is driven by soil. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. |

| | |
|---|---|
| | Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |

| | |
|---|---|
| | |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 8,7656 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,012522 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 8,7656 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,012522 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |

| ***Guidance to Downstream Users*** |
|---|
| For scaling see: http://www.ecetoc.org/tra |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>General exposures (open systems) Use domain: industrial |

| | |
|---|---|
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 24,5438 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,035063 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 24,5438 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,035063 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>General exposures (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance | 12400 Pa |

| | |
|---|---|
| during use | |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 245,4375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,350625 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 245,4375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,350625 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>General exposures (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Use suitable chemically resistant | Effectiveness: 80 % |

| | |
|---|---|
| gloves. | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 245,4375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,350625 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 245,4375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,350625 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Equipment maintenance Cleaning Use domain: industrial |
| | Additional PROC(s) covered: Manual maintenance (cleaning and repair) of machinery |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |

| | |
|---|---|
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,5486 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000272 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 7,3631 mg/m³ |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,010519 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 36,8156 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,052594 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 18,4078 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,026297 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 18,4078 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,026297 |
| ***Guidance to Downstream Users*** | |

| |
|---|
| For scaling see: http://www.ecetoc.org/tra |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 18,4078 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,026297 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 18,4078 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,026297 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Process sampling Use domain: industrial |

| | |
|---|---|
| | |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 4,1143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,002041 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 29,4525 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,042075 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 49,0875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,070125 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Process sampling Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |

| | |
|---|---|
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 4,1143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,002041 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 42,075 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,060107 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Process sampling Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 6,8571 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,003401 |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Laboratory activities Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 2. Short title of exposure scenario

ERC2; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC28, PROC8b, PROC9, PROC14, PROC15

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ESVOC SpERC 2.2.c.v2 |
| ***Operational conditions*** | |
| Annual amount per site | 1.714.000 kg |
| Minimum emission days per year | 300 |
| Emission factor air | 2,5 % |
| Emission factor water | 0,02 % |
| Emission factor soil | 0,01 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,138993 |
| | Risk from environmental exposure is driven by soil. |
| Maximum amount of safe use | 41.105,2 kg/d |
| Risk from environmental exposure is driven by soil. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |

| | |
|---|---|
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |

| | |
|---|---|
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |

| | |
|---|---|
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) with sample collection Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 8,7656 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,012522 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 8,7656 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,012522 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Batch processes at elevated temperatures Use in contained batch processes Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,4114 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000204 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 10,5188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,015027 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |

| | |
|---|---|
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Batch processes at elevated temperatures Use in contained batch processes Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>General exposures (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 6,8571 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,003401 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>General exposures (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 6,8571 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,003401 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 350,625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,500893 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 350,625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,500893 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Mixing operations (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |

| | |
|---|---|
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Equipment maintenance Cleaning Use domain: industrial |
| | Additional PROC(s) covered: Manual maintenance (cleaning and repair) of machinery |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee | Effectiveness: 90 % |

| | |
|---|---|
| training. | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>(manual) Transfer from/pouring from containers Non-dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,2286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004082 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 31,5563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,04508 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** |
|---|

| | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Drum/Batch transfers Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,2286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004082 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 31,5563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,04508 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance | 12400 Pa |

| | |
|---|---|
| during use | |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Drum/Batch transfers Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |

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|---|---|
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Drum and small package filling Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 4,1143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,002041 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 42,075 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,060107 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |

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|---|---|
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Process sampling Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 15 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 49,0875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,070125 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 490,875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,70125 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).<br>Drum and small package filling Use domain: industrial |

| | |
|---|---|
| | |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 6,8571 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,003401 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 210,375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,300536 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 210,375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,300536 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC14: Tabletting, compression, extrusion, pelletisation, granulation<br>Tabletting, compression, extrusion, pelletisation, granulation Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |

| | |
|---|---|
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 3,4286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001701 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Laboratory activities Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 3. Short title of exposure scenario

ERC6a; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC28, PROC8b, PROC9

## Control of exposure and risk management measures

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|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ERC6a: Use of intermediate |
| ***Operational conditions*** | |
| Annual amount per site | 15.000.000 kg |
| Minimum emission days per year | 300 |
| Emission factor air | 2,5 % |
| Emission factor water | 0,03 % |
| Emission factor soil | 0,1 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 40 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,593125 |
| | Risk from environmental exposure is driven by soil. |
| Maximum amount of safe use | 84.299,2 |

| | |
|---|---|
| | kg/d |
| Risk from environmental exposure is driven by soil. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Storage Use domain: industrial |

| | |
|---|---|
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |

| | |
|---|---|
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 8,7656 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,012522 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 8,7656 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,012522 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,2743 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000136 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>General exposures (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane |

| | |
|---|---|
| | Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,8229 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000408 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 14,7263 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,021038 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 24,5438 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,035063 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>General exposures (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |

| | |
|---|---|
| | |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,8229 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000408 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 14,7263 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,021038 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 24,5438 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,035063 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>General exposures (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air | Effectiveness: 30 % |

| | |
|---|---|
| changes per hour) | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 24,5438 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,035063 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 24,5438 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,035063 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Equipment maintenance Cleaning Use domain: industrial |
| | Additional PROC(s) covered: Manual maintenance (cleaning and repair) of machinery |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |

| | |
|---|---|
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,5486 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000272 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 12,2719 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,017531 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,5486 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000272 |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 7,3631 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,010519 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 36,8156 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,052594 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,5486 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000272 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 7,3631 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,010519 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 36,8156 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,052594 |
| ***Guidance to Downstream Users*** | |

| |
|---|
| For scaling see: http://www.ecetoc.org/tra |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane |

| | |
|---|---|
| | Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Process sampling Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |

| | |
|---|---|
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 4,1143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,002041 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 29,4525 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,042075 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 49,0875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,070125 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Process sampling Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 4,1143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,002041 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 42,075 mg/m³ |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,060107 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Process sampling Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 6,8571 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,003401 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 4. Short title of exposure scenario

ERC4; PROC1, PROC2, PROC3, PROC4, PROC5, PROC7, PROC8a, PROC28, PROC8b, PROC9, PROC10, PROC13, PROC14, PROC15

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ESVOC SpERC 4.4a.v1: ESVOC SpERC 4.4a.v1 |
| ***Operational conditions*** | |
| Annual amount per site | 7.900.000 kg |
| Minimum emission days per year | 300 |
| Emission factor air | 98 % |
| Emission factor water | 0,07 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Air treatment measures considered suitable are, e.g. | Wet scrubber - for dusts, Filtration, Waste gas treatment by thermal oxidation, Adsorption |
| Wastewater treatment measures considered suitable are, e.g. | Acclimated biological treatment, Distillation |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,595957 |
| | Risk from environmental exposure is driven by soil. |
| Maximum amount of safe use | 44.186,6 kg/d |
| Risk from environmental exposure is driven by soil. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. |

| | |
|---|---|
| | General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |

| | |
|---|---|
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |

| | |
|---|---|
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Film formation - force drying, stoving and other technologies Use in contained systems elevated temperature Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 8,7656 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,012522 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |

| | |
|---|---|
| Exposure estimate | 8,7656 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,012522 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) Use in contained systems with sample collection Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Storage Use domain: industrial |

| | |
|---|---|
| | |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) Use in contained systems with sample collection Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |

| | |
|---|---|
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Mixing operations Use in contained batch processes Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |

| | |
|---|---|
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Film formation - air drying (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 4,1143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,002041 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 21,0375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,030054 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |

| ***Guidance to Downstream Users*** |
|---|
| For scaling see: http://www.ecetoc.org/tra |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Film formation - air drying (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 6,8571 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,003401 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Preparation of material for application Mixing operations (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane |

| | |
|---|---|
| | Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,2286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004082 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes Preparation of material for application Mixing operations (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |

| | |
|---|---|
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC7: Industrial spraying<br>Spraying (manual) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Wear chemically resistant gloves in combination with specific activity training | Effectiveness: 95 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |

| | |
|---|---|
| Exposure estimate | 1,2857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000638 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 31,5563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,04508 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC7: Industrial spraying<br>Spraying (automatic/robotic) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,5714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004252 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Equipment maintenance Cleaning Use domain: industrial |
| | Additional PROC(s) covered: Manual maintenance (cleaning and repair) of machinery |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Material transfers Non-dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,2286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004082 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Material transfers Non-dedicated facility Use domain: industrial |

| | |
|---|---|
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Material transfers Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |

| | |
|---|---|
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,2286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004082 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 31,5563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,04508 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Material transfers Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Material transfers Drum/Batch transfers Transfer from/pouring from containers Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 4,1143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,002041 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 42,075 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,060107 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Material transfers Drum/Batch transfers Transfer from/pouring from containers Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 6,8571 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,003401 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC10: Roller application or brushing<br>Roller, spreader, flow application Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Wear chemically resistant gloves in combination with specific activity training | Effectiveness: 95 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,8229 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000408 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC10: Roller application or brushing<br>Roller, spreader, flow application Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |

| | |
|---|---|
| Wear chemically resistant gloves in combination with specific activity training | Effectiveness: 95 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring. Dipping, immersion and pouring Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,2286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004082 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring. Dipping, immersion and pouring Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC14: Tabletting, compression, extrusion, pelletisation, |

| | |
|---|---|
| | granulation<br>Tabletting, compression, extrusion, pelletisation, granulation Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 3,4286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001701 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Laboratory activities Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |

| | |
|---|---|
| | |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 5. Short title of exposure scenario

ERC4; PROC1, PROC2, PROC3, PROC4, PROC7, PROC8a, PROC8b, PROC10, PROC13

## Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ESVOC SpERC 4.6a.v1: ESVOC SpERC 4.6a.v1 |
| ***Operational conditions*** | |
| Annual amount per site | 190.000 kg |
| Minimum emission days per year | 20 |
| Emission factor air | 30 % |
| Emission factor water | 3 ppm |
| Emission factor soil | 0 % |
| | Values provided in per mill |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |

| | |
|---|---|
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Air treatment measures considered suitable are, e.g. | Wet scrubber - for dusts, Waste gas treatment by thermal oxidation, Adsorption |
| Wastewater treatment measures considered suitable are, e.g. | Acclimated biological treatment, Distillation |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,006817 |
| | Risk from environmental exposure is driven by freshwater. |
| Maximum amount of safe use | 1.393,5 t/d |
| Risk from environmental exposure is driven by freshwater. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Automated process with (semi) closed systems. Use in contained systems Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 25 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,8229 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000408 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 36,8156 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,052594 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 36,8156 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,052594 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Automated process with (semi) closed systems. Use in contained systems Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 25 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,8229 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000408 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Application of cleaning products in closed systems Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |

| | |
|---|---|
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Application of cleaning products in closed systems Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Automated process with (semi) closed systems. Use in contained systems Drum/Batch transfers Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,1371 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** |
|---|

| | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use in contained batch processes elevated temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 21,0375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,030054 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 21,0375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,030054 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC7: Industrial spraying<br>Cleaning with high pressure washers Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 1 % |

| | |
|---|---|
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Wear chemically resistant gloves in combination with specific activity training | Effectiveness: 95 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,2143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000106 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Bulk transfer Non-dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |

| | |
|---|---|
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,2286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004082 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Bulk transfer Non-dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,2286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004082 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 31,5563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,04508 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Filling / preparation of equipment from drums or containers. Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |

| | |
|---|---|
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,5486 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000272 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC10: Roller application or brushing<br>Cleaning with low-pressure washers Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Wear chemically resistant gloves in | Effectiveness: 90 % |

| | |
|---|---|
| combination with ‘basic’ employee training. | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,5486 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000272 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC10: Roller application or brushing<br>(manual) Cleaning Surfaces no spraying Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,5486 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000272 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring. Degreasing small objects in cleaning station Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,2286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004082 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring. Degreasing small objects in cleaning station Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 6. Short title of exposure scenario

ERC4; PROC1, PROC2, PROC3, PROC8b, PROC9, PROC12

## Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ESVOC SpERC 6.1a.v1: ESVOC SpERC 6.1a.v1 |
| ***Operational conditions*** | |
| Annual amount per site | 1.714.000 kg |
| Minimum emission days per year | 300 |
| Emission factor air | 100 % |
| Emission factor water | 0,003 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Air treatment measures considered suitable are, e.g. | Waste gas treatment by thermal oxidation |
| Wastewater treatment measures considered suitable are, e.g. | Acclimated biological treatment |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,527927 |
| | Risk from environmental exposure is driven by soil. |
| Maximum amount of safe use | 10.822,2 kg/d |
| Risk from environmental exposure is driven by soil. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Mixing operations elevated temperature Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Mixing operations elevated temperature Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |

| | |
|---|---|
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Drying and storage Use in contained systems Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Drying and storage Use in contained systems Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional |

| | |
|---|---|
| | controlled exposure or processes with equivalent containment condition<br>Centrifuging including discharging Use in contained batch processes Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Semi-bulk packaging Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,2286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004082 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 31,5563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,04508 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Semi-bulk packaging Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |

| | |
|---|---|
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 8,2286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,004082 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 220,8938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,315563 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 368,1563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,525938 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Drum and small package filling Filling / preparation of equipment from drums or containers. Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 4,1143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,002041 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 42,075 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,060107 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Drum and small package filling Filling / preparation of equipment from drums or containers. Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 6,8571 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,003401 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC12: Use of blowing agents in manufacture of foam Extrusion and expansion of polymer mass Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,2057 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000102 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 21,0375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,030054 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC12: Use of blowing agents in manufacture of foam Production of foam-based objects Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |

| | |
|---|---|
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,2057 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000102 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 21,0375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,030054 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC12: Use of blowing agents in manufacture of foam Extrusion and expansion of polymer mass Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC12: Use of blowing agents in manufacture of foam Production of foam-based objects Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC12: Use of blowing agents in manufacture of foam Treatment by heating elevated temperature Use domain: industrial |
| ***Operational conditions*** | |

| | |
|---|---|
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC12: Use of blowing agents in manufacture of foam Cutting by heated wire Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |

| | |
|---|---|
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 7. Short title of exposure scenario

ERC4; PROC1, PROC2, PROC3, PROC4, PROC6, PROC7, PROC8a, PROC28, PROC8b, PROC10, PROC13, PROC14

## Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ESVOC SpERC 4.20.v1: ESVOC SpERC 4.20.v1 |
| ***Operational conditions*** | |
| Annual amount per site | 7.900.000 kg |
| Minimum emission days per year | 100 |
| Emission factor air | 20 % |
| Emission factor water | 3 ppm |
| Emission factor soil | 0 % |
| | Values provided in per mill |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |

| | |
|---|---|
| | |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Air treatment measures considered suitable are, e.g. | Waste gas treatment by thermal oxidation, Adsorption |
| Wastewater treatment measures considered suitable are, e.g. | Acclimated biological treatment, Distillation |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,018598 |
| | Risk from environmental exposure is driven by soil. |
| Maximum amount of safe use | 4.247,7<br>t/d |
| Risk from environmental exposure is driven by soil. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Material transfers (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** |
|---|

| | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Material transfers (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Material transfers (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance | 12400 Pa |

| | |
|---|---|
| during use | |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 8,7656 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,012522 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 8,7656 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,012522 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Material transfers (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - local |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Mixing operations Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises |

| | |
|---|---|
| | Mixing operations (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,8229 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000408 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 21,0375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,030054 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Mixing operations (open systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |

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|---|---|
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC6: Calendering operations<br>Casting operations (open systems) elevated temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 25 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Wear chemically resistant gloves in | Effectiveness: 90 % |

| | |
|---|---|
| combination with ‘basic’ employee training. | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,9874 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00049 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 94,6688 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,135241 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 157,7813 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,225402 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC6: Calendering operations<br>Casting operations (open systems) elevated temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |

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|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,6457 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000816 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 157,7813 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,225402 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 262,9688 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,37567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC7: Industrial spraying<br>Spraying Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Wear chemically resistant gloves in combination with specific activity training | Effectiveness: 95 % |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,1429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001063 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |

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|---|---|
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC7: Industrial spraying<br>Spraying (manual) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 5,1429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,002551 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 31,5563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,04508 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |

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| For scaling see: http://www.ecetoc.org/tra |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Equipment maintenance Cleaning Use domain: industrial |
| | Additional PROC(s) covered: Manual maintenance (cleaning and repair) of machinery |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** |
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|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Drum/Batch transfers Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Drum/Batch transfers Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,6457 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000816 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 31,5563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,04508 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC10: Roller application or brushing<br>(manual) Rolling/ Brushing Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |

| | |
|---|---|
| Wear chemically resistant gloves in combination with specific activity training | Effectiveness: 95 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,8229 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000408 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC10: Roller application or brushing<br>(manual) Rolling/ Brushing Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Wear chemically resistant gloves in combination with specific activity training | Effectiveness: 95 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |

| | |
|---|---|
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring. Treatment by dipping and pouring Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,6457 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000816 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 36,8156 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,052594 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring. Treatment by dipping and pouring Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** |
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|---|---|
| **Use descriptors covered** | PROC14: Tabletting, compression, extrusion, pelletisation, granulation<br>Mold forming Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,4114 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000204 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC14: Tabletting, compression, extrusion, pelletisation, granulation<br>Mold forming Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |

| | |
|---|---|
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 8. Short title of exposure scenario

ERC7; PROC1, PROC2, PROC3, PROC8a, PROC28, PROC8b, PROC16

## Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ESVOC SpERC 7.12a.a.v3 |
| ***Operational conditions*** | |
| Annual amount per site | 190.000 kg |
| Minimum emission days per year | 300 |
| Emission factor air | 5 % |
| Emission factor water | 0,001 % |

| | |
|---|---|
| | |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,009222 |
| | Risk from environmental exposure is driven by freshwater. |
| Maximum amount of safe use | 68.677,4<br>kg/d |
| Risk from environmental exposure is driven by freshwater. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>General exposures (closed systems) With occasional controlled exposure. Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>General exposures (closed systems) With occasional controlled exposure. Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any | |

| | |
|---|---|
| skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities |

| | |
|---|---|
| | Equipment maintenance Use domain: industrial |
| | Additional PROC(s) covered: Manual maintenance (cleaning and repair) of machinery |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 95 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Drum/Batch transfers Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |

| | |
|---|---|
| Local exhaust ventilation | Effectiveness: 95 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 26,2969 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,037567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC16: Use of fuels<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |

| | |
|---|---|
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 9. Short title of exposure scenario

ERC6c; PROC1, PROC2, PROC3, PROC4, PROC8b, PROC21

## Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ESVOC SpERC 2.2.v1: ESVOC SpERC 2.2.v1 |
| ***Operational conditions*** | |
| Annual amount per site | 7.900.000 kg |
| Minimum emission days per year | 300 |
| Emission factor air | 0,01 % |
| Emission factor water | 0,001 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Air treatment measures considered suitable are, e.g. | Adsorption |
| Wastewater treatment measures considered suitable are, e.g. | Acclimated biological treatment |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,011296 |
| | Risk from environmental exposure is driven by freshwater. |
| Maximum amount of safe use | 2.331,2 |

| | |
|---|---|
| | t/d |
| Risk from environmental exposure is driven by freshwater. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Continuous process no sampling Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Polymerisation (bulk and batch) Continuous process with sample collection Use domain: industrial |

| | |
|---|---|
| | |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage With occasional controlled exposure. Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |

| | |
|---|---|
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Polymerisation (bulk and batch) Batch process with sample collection Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |

| | |
|---|---|
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Polymerisation (bulk and batch) Batch process with sample collection Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Finishing operations Batch process with sample collection<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition |

| | |
|---|---|
| | Additivation and stabilisation Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Bulk transfer Continuous process with sample collection<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |

| | |
|---|---|
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Intermediate polymer storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 6,8571 mg/kg bw/day |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,003401 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 350,625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,500893 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 350,625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,500893 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer transport with sample collection Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Pelletisation and pellet screening (open systems) Rework of articles Use domain: industrial |
| | Additional PROC(s) covered: Low energy manipulation and handling of substances bound in/on materials or articles |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>transport with sample collection Use domain: industrial |
| ***Operational conditions*** | |

| | |
|---|---|
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 13,7143 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,006803 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 10. Short title of exposure scenario

ERC4; PROC1, PROC2, PROC3, PROC4, PROC5, PROC6, PROC8a, PROC28, PROC8b, PROC13, PROC14, PROC21

## Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ESVOC SPERC 4.24.v1: ESVOC SPERC 4.24.v1 |
| ***Operational conditions*** | |
| Annual amount per site | 7.900.000 kg |
| Minimum emission days per year | 300 |
| Emission factor air | 15 % |

| | |
|---|---|
| | |
| Emission factor water | 0 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Air treatment measures considered suitable are, e.g. | Wet scrubber - for dusts, Waste gas treatment by thermal oxidation, Adsorption |
| Wastewater treatment measures considered suitable are, e.g. | Acclimated biological treatment, Distillation |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,037379 |
| | Risk from environmental exposure is driven by soil. |
| Maximum amount of safe use | 704.492,7 kg/d |
| Risk from environmental exposure is driven by soil. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Bulk transfer (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |

| | |
|---|---|
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Bulk weighing (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0245 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000035 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Bulk transfer (closed systems) Use domain: industrial |

| | |
|---|---|
| | |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Bulk weighing (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |

| | |
|---|---|
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,0351 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00005 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Bulk transfer (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Bulk weighing (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** |
|---|

| | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Bulk transfer (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Bulk weighing (closed systems) Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance | 12400 Pa |

| | |
|---|---|
| during use | |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Additive premixing Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Additive premixing Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 245,4375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,350625 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 245,4375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,350625 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Additive premixing Use domain: industrial |

| | |
|---|---|
| | |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 262,9688 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,37567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 262,9688 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,37567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC6: Calendering operations<br>Calendering (including Banburys) elevated temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |

| | |
|---|---|
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Wear chemically resistant gloves in combination with 'basic' employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,6457 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000816 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC6: Calendering operations<br>Calendering (including Banburys) elevated temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |

| | |
|---|---|
| Wear chemically resistant gloves in combination with 'basic' employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Equipment maintenance Use domain: industrial |
| | Additional PROC(s) covered: Manual maintenance (cleaning and repair) of machinery |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |

| | |
|---|---|
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 87,6563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,125223 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Bulk transfer Dedicated facility Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring. Production of articles by dipping and pouring Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 25 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,6457 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000816 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC14: Tabletting, compression, extrusion, pelletisation, granulation |

| | |
|---|---|
| | Extrusion and masterbatching Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 3,4286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001701 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 262,9688 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,37567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 262,9688 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,37567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC14: Tabletting, compression, extrusion, pelletisation, granulation<br>Injection moulding (of articles) Use domain: industrial |
| | Additional PROC(s) covered: Low energy manipulation and handling of substances bound in/on materials or articles |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 3,4286 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001701 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 262,9688 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,37567 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 262,9688 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,37567 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 11. Short title of exposure scenario

ERC4; PROC15

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article) |
| ***Operational conditions*** | |
| Annual amount per site | 5.000 kg |
| Minimum emission days per year | 20 |

| | |
|---|---|
| Emission factor air | 1 % |
| Emission factor water | 0,1 % |
| Emission factor soil | 0,1 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,554272 |
| | Risk from environmental exposure is driven by soil. |
| Maximum amount of safe use | 451<br>kg/d |
| Risk from environmental exposure is driven by soil. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Laboratory activities Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0686 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 24,5438 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,035063 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Laboratory activities small scale Use domain: industrial |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 12. Short title of exposure scenario

ERC8d; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC28, PROC8b, PROC10, PROC11, PROC13, PROC15, PROC19

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ESVOC SpERC 8.3c.v2 |
| ***Operational conditions*** | |
| Annual amount used in the EU | 2.700.000 kg |
| Minimum emission days per year | 365 |
| Emission factor air | 98,5 % |
| Emission factor water | 1 % |
| Emission factor soil | 0,5 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,006813 |
| | Risk from environmental exposure is driven by freshwater. |
| Maximum amount of safe use | 217,2<br>kg/d |
| Risk from environmental exposure is driven by freshwater. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: professional |

| | |
|---|---|
| | |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Storage Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |

| | |
|---|---|
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,3506 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000501 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,3506 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000501 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Filling / preparation of equipment from drums or containers. Use in contained systems Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,8229 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000408 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 7,3631 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,010519 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 12,2719 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,017531 |

| ***Guidance to Downstream Users*** |
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| For scaling see: http://www.ecetoc.org/tra |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Filling / preparation of equipment from drums or containers. Use in contained systems Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,8229 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000408 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 10,5188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,015027 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 17,5313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,025045 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) Use in contained |

| | |
|---|---|
| | systems Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Filling / preparation of equipment from drums or containers. Use in contained systems Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance | 12400 Pa |

| | |
|---|---|
| during use | |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Preparation of material for application Use in contained batch processes Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,1371 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

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|---|---|
| | Worker - inhalation, long-term - local |
| Exposure estimate | 350,625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,500893 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Roller, spreader, flow application Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 25 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0823 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000041 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Preparation of material for application Use domain: |

| | |
|---|---|
| | professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,6457 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000816 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 73,6313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,105188 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes Preparation of material for application Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,6457 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000816 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 73,6313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,105188 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes Preparation of material for application Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |

| | |
|---|---|
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,6457 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000816 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 73,6313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,105188 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes Preparation of material for application Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,6457 mg/kg bw/day |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,000816 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Material transfers Drum/Batch transfers Non-dedicated facility Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,5486 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000272 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 24,5438 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,035063 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |

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|---|---|
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Equipment maintenance Cleaning Use domain: professional |
| | Additional PROC(s) covered: Manual maintenance (cleaning and repair) of machinery |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 80 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,5486 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000272 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 49,0875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,070125 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 245,4375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,350625 |
| ***Guidance to Downstream Users*** | |

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|---|
| For scaling see: http://www.ecetoc.org/tra |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Material transfers Drum/Batch transfers Dedicated facility<br>Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,6457 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000816 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 36,8156 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,052594 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities |

| | |
|---|---|
| | Material transfers Drum/Batch transfers Dedicated facility<br>Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC10: Roller application or brushing<br>Roller, spreader, flow application Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 3,2914 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001633 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 73,6313 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,105188 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC11: Non industrial spraying<br>Spraying (manual) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Wear chemically resistant gloves in | Effectiveness: 90 % |

| | |
|---|---|
| combination with ‘basic’ employee training. | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,2857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000638 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 29,4525 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,042075 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 49,0875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,070125 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC11: Non industrial spraying<br>Spraying (manual) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,2857 mg/kg bw/day |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,000638 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 42,075 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,060107 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC11: Non industrial spraying<br>Spraying (manual) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 80 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,1429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001063 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 140,25 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,200357 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 140,25 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,200357 |

| ***Guidance to Downstream Users*** |
|---|
| For scaling see: http://www.ecetoc.org/tra |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring. Dipping, immersion and pouring Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 61,3594 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,087656 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring. Dipping, immersion and pouring Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane |

| | |
|---|---|
| | Content: >= 0 % - <= 25 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 80 % |
| Wear chemically resistant gloves in combination with 'basic' employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,4937 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000245 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 63,1125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,090161 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Laboratory activities Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |

| | |
|---|---|
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC19: Manual activities involving hand contact<br>Hand application - fingerpaints, pastels, adhesives Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,5657 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000281 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 21,0375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,030054 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC19: Manual activities involving hand contact<br>Hand application - fingerpaints, pastels, adhesives Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,5657 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000281 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 49,0875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,070125 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |

| | |
|---|---|
| Exposure estimate | 245,4375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,350625 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 13. Short title of exposure scenario

ERC8d; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC10, PROC11, PROC13, PROC19

## Control of exposure and risk management measures

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|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ESVOC SpERC 8.4b.v3 |
| ***Operational conditions*** | |
| Annual amount used in the EU | 190.000 kg |
| Minimum emission days per year | 365 |
| Emission factor air | 4 % |
| Emission factor water | 1 ppm |
| Emission factor soil | 0,2 ppm |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,006366 |
| | Risk from environmental exposure is driven by freshwater. |
| Maximum amount of safe use | 16,4<br>kg/d |
| Risk from environmental exposure is driven by freshwater. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent |

| | |
|---|---|
| | containment conditions.<br>Storage Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,3506 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000501 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,3506 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000501 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Automated process with (semi) closed systems. Use in contained systems Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |

| | |
|---|---|
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 52,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,075134 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Automated process with (semi) closed systems. Use in contained systems Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |

| | |
|---|---|
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |

| | |
|---|---|
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Automated process with (semi) closed systems. Use in contained systems Drum/Batch transfers Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,1371 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 70,125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,100179 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 350,625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,500893 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for |

| | |
|---|---|
| | exposure arises<br>Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,1371 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Batch process (open systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |

| | |
|---|---|
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| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,2743 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000136 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Cleaning of medical devices Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |

| | |
|---|---|
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 6,8571 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,003401 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 613,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,876563 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 613,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,876563 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Filling / preparation of equipment from drums or containers. Non-dedicated facility Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 25 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |

| | |
|---|---|
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3291 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000163 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 21,0375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,030054 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Filling / preparation of equipment from drums or containers. Dedicated facility Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 25 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3291 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000163 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 105,1875 mg/m³ |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,150268 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 525,9375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,751339 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC10: Roller application or brushing<br>Cleaning with low-pressure washers Rolling/ Brushing no spraying Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,0971 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000544 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC10: Roller application or brushing<br>(manual) Surfaces Cleaning Spraying Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,0971 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000544 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC11: Non industrial spraying<br>Cleaning with high pressure washers Spraying Use domain: professional |
| ***Operational conditions*** | |

| | |
|---|---|
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 1 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,2857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000638 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 6,3113 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,009016 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 10,5188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,015027 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC11: Non industrial spraying<br>Cleaning with high pressure washers Spraying Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 1 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |

| | |
|---|---|
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,2857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000638 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 14,7263 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,021038 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 24,5438 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,035063 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring. (manual) Cleaning Surfaces Dipping, immersion and pouring Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 25 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Use suitable chemically resistant | Effectiveness: 80 % |

| | |
|---|---|
| gloves. | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3291 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000163 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 31,5563 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,04508 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 157,7813 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,225402 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC19: Manual activities involving hand contact<br>Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour) | Effectiveness: 70 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |

| | |
|---|---|
| Exposure estimate | 1,6971 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000842 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 63,1125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,090161 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 14. Short title of exposure scenario

ERC9b; PROC1, PROC2, PROC3, PROC8a, PROC28, PROC16

## Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ERC9b: Widespread use of functional fluid (outdoor) |
| ***Operational conditions*** | |
| Annual amount used in the EU | 190.000 kg |
| Minimum emission days per year | 365 |
| Emission factor air | 5 % |
| Emission factor water | 5 % |
| Emission factor soil | 5 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,006523 |
| | Risk from environmental exposure is driven by freshwater. |

| | |
|---|---|
| Maximum amount of safe use | 159,6<br>kg/d |
| Risk from environmental exposure is driven by freshwater. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Storage Use domain: professional |

| | |
|---|---|
| | |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>General exposures (closed systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |

| | |
|---|---|
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,3506 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000501 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,3506 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000501 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 60 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,2743 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000136 |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>General exposures (closed systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>(closed systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane |

| | |
|---|---|
| | Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,6857 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00034 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 350,625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,500893 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 350,625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,500893 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Equipment maintenance Use domain: professional |
| | Additional PROC(s) covered: Manual maintenance (cleaning and repair) of machinery |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |

| | |
|---|---|
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 80 % |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Wear chemically resistant gloves in combination with 'basic' employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,8229 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000408 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 21,0375 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,030054 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC16: Use of fuels<br>(closed systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 15. Short title of exposure scenario

ERC8a; PROC15

### Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ESVOC SpERC 8.17.v2 |
| ***Operational conditions*** | |
| Annual amount used in the EU | 286.000 kg |
| Minimum emission days per year | 365 |
| Emission factor air | 50 % |
| Emission factor water | 50 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,00873 |
| | Risk from environmental exposure is driven by freshwater. |
| Maximum amount of safe use | 18 |

| | |
|---|---|
| | kg/d |
| Risk from environmental exposure is driven by freshwater. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Laboratory activities small scale Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 35,0625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,050089 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Laboratory activities Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |

| | |
|---|---|
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,3429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 16. Short title of exposure scenario

ERC8d; PROC1, PROC2, PROC6, PROC21, PROC8a, PROC28, PROC8b

### Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ESVOC SpERC 8.21b.v2 |
| ***Operational conditions*** | |
| Annual amount used in the EU | 2.700.000 kg |
| Minimum emission days per year | 365 |
| Emission factor air | 98 % |

| | |
|---|---|
| Emission factor water | 1 % |
| Emission factor soil | 1 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,006813 |
| | Risk from environmental exposure is driven by freshwater. |
| Maximum amount of safe use | 217,2<br>kg/d |
| Risk from environmental exposure is driven by freshwater. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Bulk transfer (closed systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Storage Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,2454 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000351 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions.<br>Bulk transfer (closed systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 0,0343 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000017 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 0,3506 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000501 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 0,3506 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000501 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Bulk transfer (closed systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |

| | |
|---|---|
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Outdoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 122,7188 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,175313 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Bulk transfer (closed systems) Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions<br>Storage Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC6: Calendering operations<br>Injection moulding (of articles) Use domain: professional |
| | Additional PROC(s) covered: Low energy manipulation and handling of substances bound in/on materials or articles |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 240 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Wear suitable respiratory protection. | Effectiveness: 90 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,6457 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000816 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 105,1875 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,150268 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 175,3125 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,250446 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| |
|---|
| **Contributing exposure scenario** |

| | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Equipment maintenance Use domain: professional |
| | Additional PROC(s) covered: Manual maintenance (cleaning and repair) of machinery |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Local exhaust ventilation | Effectiveness: 80 % |
| Wear chemically resistant gloves in combination with ‘basic’ employee training. | Effectiveness: 90 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 1,3714 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,00068 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 350,625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,500893 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 350,625 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,500893 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Material transfers Dedicated facility Use domain: professional |

| | |
|---|---|
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 613,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,876563 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 613,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,876563 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Material transfers Dedicated facility Use domain: professional |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 100 % |
| Physical state | liquid |
| Vapour pressure of the substance | 12400 Pa |

| | |
|---|---|
| during use | |
| Duration and Frequency of activity | 480 min 5 days per week |
| Indoor/Outdoor | Indoor |
| | Assumes activities are at ambient temperature. |
| ***Risk Management Measures*** | |
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) | Effectiveness: 30 % |
| Use suitable chemically resistant gloves. | Effectiveness: 80 % |
| Avoid contact with eyes. Wash off any skin contamination immediately. | |
| Use suitable eye protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - dermal, long-term - systemic |
| Exposure estimate | 2,7429 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001361 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - systemic |
| Exposure estimate | 613,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,876563 |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Worker |
| | Worker - inhalation, long-term - local |
| Exposure estimate | 613,5938 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,876563 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 17. Short title of exposure scenario

ERC9b; PC13

## Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ERC9b: Widespread use of functional fluid (outdoor) |
| ***Operational conditions*** | |
| Annual amount used in the EU | 190.000 kg |
| Minimum emission days per year | 365 |
| Emission factor air | 5 % |

| | |
|---|---|
| Emission factor water | 5 % |
| Emission factor soil | 5 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,006523 |
| | Risk from environmental exposure is driven by freshwater. |
| Maximum amount of safe use | 159,6<br>kg/d |
| Risk from environmental exposure is driven by freshwater. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PC13: Fuels. |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 3 min 365 uses per year |
| Indoor/Outdoor | Outdoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 0,2 % |
| | Amount per use 37.500 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 1,3543 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001142 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |

| | |
|---|---|
| Exposure estimate | 0,0734 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000357 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PC13: Fuels. |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 1,98 min 365 uses per year |
| Indoor/Outdoor | Outdoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 2 % |
| | Amount per use 3.750 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 1,3543 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,001142 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 0,0495 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,00024 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PC13: Fuels. |
| ***Operational conditions*** | |

| | |
|---|---|
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 2 h 365 uses per year |
| Indoor/Outdoor | Outdoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction inhalation | 2 % |
| | Amount per use 750 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 0,1242 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000603 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PC13: Fuels. |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 1,8 min 365 uses per year |
| Indoor/Outdoor | Indoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 3 % |
| | Amount per use 750 g Relevant for inhalative exposure estimates |

| ***Exposure estimate and reference to its source*** | |
|---|---|
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,5417 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000457 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 0,0404 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000196 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

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## 18. Short title of exposure scenario

ERC8d; PC1, PC9a, PC15, PC18, PC24, PC34

## Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ESVOC SpERC 8.3b.v2 |
| ***Operational conditions*** | |
| Annual amount used in the EU | 190.000 kg |
| Minimum emission days per year | 365 |
| Emission factor air | 98 % |
| Emission factor water | 1 % |
| Emission factor soil | 1 % |
| Receive Surf. Water (Flow Rate). | 18.000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2.000 m3/d |
| ***Exposure estimate and reference to its source*** | |

| | |
|---|---|
| Assessment method | EASY TRA v5.2, ECETOC TRA v3.0, Environment |
| Risk Characterization Ratio (RCR) | 0,006398 |
| | Risk from environmental exposure is driven by freshwater. |
| Maximum amount of safe use | 16,3<br>kg/d |
| Risk from environmental exposure is driven by freshwater. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PC1: Adhesives, Sealants |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 0,1 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 6 h 12 uses per year |
| Indoor/Outdoor | Indoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 6.390 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 21,5813 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,104763 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PC1: Adhesives, Sealants |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane |

| | |
|---|---|
| | Content: >= 0 % - <= 10 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 4 h 12 uses per year |
| Indoor/Outdoor | Indoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction inhalation | 100 % |
| | Amount per use 85,05 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0184 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000016 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool), Inhalation model: Exposure to spray/dust |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 26,8522 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,130351 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PC9a: Coatings and paints, thinners, paint removers |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 0,25 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 2,2 h 12 uses per year |
| Indoor/Outdoor | Indoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 2.760 g Relevant for inhalative exposure estimates |

| ***Exposure estimate and reference to its source*** | |
|---|---|
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0055 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000005 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 17,5582 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,085234 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PC9a: Coatings and paints, thinners, paint removers |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 1 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 2,2 h 12 uses per year |
| Indoor/Outdoor | Indoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 744 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0221 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000019 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 18,9323 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,091905 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |

| ***Guidance to Downstream Users*** |
|---|
| For scaling see: http://www.ecetoc.org/tra |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PC9a: Coatings and paints, thinners, paint removers |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 0,1 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 19,8 min 12 uses per year |
| Indoor/Outdoor | Indoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 215 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool), Inhalation model: Exposure to spray/dust |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 0,0686 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000333 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PC9a: Coatings and paints, thinners, paint removers |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 2 % |
| Vapour pressure of the substance during use | 12400 Pa |

| | |
|---|---|
| Duration and Frequency of activity | Exposure duration: 2 h 12 uses per year |
| Indoor/Outdoor | Indoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 491 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0885 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000075 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 23,8273 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,115667 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PC15: Non-metal-surface treatment products. |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 0,25 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 2,2 h 12 uses per year |
| Indoor/Outdoor | Indoor |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 2.760 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0055 mg/kg bw/day |

| | |
|---|---|
| Risk Characterization Ratio (RCR) | 0,000005 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 17,5582 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,085234 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PC15: Non-metal-surface treatment products. |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 1 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 2,2 h 12 uses per year |
| Indoor/Outdoor | Indoor |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 744 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0221 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000019 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 18,9323 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,091905 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PC15: Non-metal-surface treatment products. |

| | |
|---|---|
| | |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 5 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 19,8 min 12 uses per year |
| Indoor/Outdoor | Indoor |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 215 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool), Inhalation model: Exposure to spray/dust |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 3,429 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,016646 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PC15: Non-metal-surface treatment products. |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 2 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 2 h 12 uses per year |
| Indoor/Outdoor | Indoor |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |

| | |
|---|---|
| | |
| | Amount per use 491 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0885 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000075 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 23,8273 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,115667 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PC18: Ink and Toners. |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 10 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 2,2 h 365 uses per year |
| Indoor/Outdoor | Indoor |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 40 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,9209 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000777 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 10,1787 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,049411 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |

| | |
|---|---|
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PC24: Lubricants, Greases and Release Products |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 10 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 10,2 min 12 uses per year |
| Indoor/Outdoor | Indoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 1 % |
| | Amount per use 2.200 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,2415 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000204 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 0,4046 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,001964 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PC24: Lubricants, Greases and Release Products |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 10 % |
| Vapour pressure of the substance | 12400 Pa |

| | |
|---|---|
| during use | |
| Duration and Frequency of activity | Exposure duration: 1 h 12 uses per year |
| Indoor/Outdoor | Indoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 34 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,2415 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000204 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 0,0001 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,000001 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PC24: Lubricants, Greases and Release Products |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 10 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 10,2 min 12 uses per year |
| Indoor/Outdoor | Indoor |
| Exposed skin area | Both hands (820 cm²) |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 73 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |

| | |
|---|---|
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,2212 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000187 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool), Inhalation model: Exposure to spray/dust |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 2,4579 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,011932 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |
| For scaling see: http://www.ecetoc.org/tra | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PC34: Textile dyes, finishing and impregnating products: icluding bleaches and other processing aids |
| ***Operational conditions*** | |
| Concentration of the substance | cyclohexane<br>Content: >= 0 % - <= 1 % |
| Vapour pressure of the substance during use | 12400 Pa |
| Duration and Frequency of activity | Exposure duration: 1 h 365 uses per year |
| Indoor/Outdoor | Indoor |
| Uptake fraction dermal | 100 % |
| Uptake fraction inhalation | 100 % |
| | Amount per use 115 g Relevant for inhalative exposure estimates |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - dermal, long-term - systemic |
| Exposure estimate | 0,0111 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000009 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - inhalation, long-term - systemic |
| Exposure estimate | 1,8016 mg/m³ |
| Risk Characterization Ratio (RCR) | 0,008746 |
| Assessment method | EASY TRA v5.2, Other consideration (non-standard tool) |
| | Consumer - oral, long-term - systemic |
| Exposure estimate | 0,0001 mg/kg bw/day |
| Risk Characterization Ratio (RCR) | 0,000001 |
| ***Guidance to Downstream Users*** | |

For scaling see: http://www.ecetoc.org/tra

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